

Chapter 6a: Degree Theory and Differential Calculus in Banach Spaces

Pathak: Introduction to Nonlinear Analysis and Fixed Point Theory

Lecture Notes

July 14, 2026

- **Section 6.1:** The Topological Degree
 - Axioms and definition in finite dimensions
 - Brouwer degree theory
 - Leray-Schauder degree for infinite dimensions
- **Section 6.2:** Degree Theory of Completely Continuous Vector Fields
 - Extension to Banach spaces
 - Homotopy properties and invariance
 - Applications to operator equations and ODEs
- **Key Goals:**
 - Extend fixed point theory to infinite dimensions
 - Prove existence of solutions without explicit computation
 - Study properties preserved under continuous deformations

6.1.0: Historical Context and Motivation

- **Brouwer (1912):** Introduced the concept of local degree in $\mathbb{R}^n$
- **Hadamard:** Developed theory and provided detailed discussion
- **Brouwer:** Extended to large degree and proved many famous fixed point results
- **Leray-Schauder:** Simple construction extending degree to infinite dimensions
- **Applications:** Existence results for nonlinear operator equations without explicit solutions

Central Question: For equation $f(x) = y$ where $f : \Omega \rightarrow \mathbb{R}^n$, can we determine number of solutions?

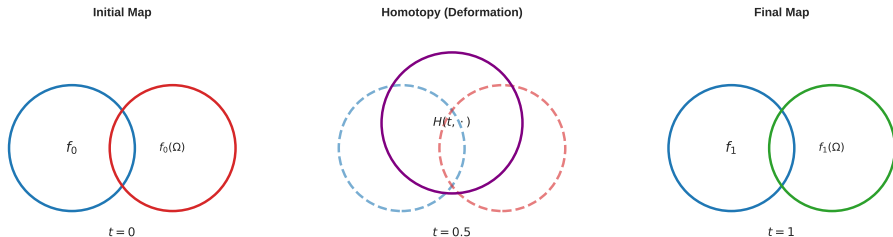
6.1.1: Basic Concepts

Definition 6.1: (Homotopy) If $f_1, f_2 : X \rightarrow Y$ are continuous maps, a *homotopy* from f_1 to f_2 is a continuous map

$$H : [0, 1] \times X \rightarrow Y \quad \text{such that} \quad H(0, x) = f_1(x), \quad H(1, x) = f_2(x)$$

Write $f_1 \simeq f_2$ if such a homotopy exists.

Key Insight: Homotopy formalizes the idea of continuous deformation.



6.1.1: Axioms of Brouwer Degree

Setup: $\Omega \subseteq \mathbb{R}^n$ open and bounded, $f : \Omega \rightarrow \mathbb{R}^n$ continuous, $y \in \mathbb{R}^n$.

Definition 6.2: The degree $\deg(f, \Omega, y)$ is an integer assigned to each admissible triple (f, Ω, y) such that:

- ① **Existence:** $\deg(f, \Omega, y) \neq 0 \Rightarrow \exists x \in \Omega : f(x) = y$
- ② **Homotopy Invariance:** If $H : [0, 1] \times \Omega \rightarrow \mathbb{R}^n$ continuous with $H(t, x) \neq y$ on $\partial\Omega$ and all $t \in [0, 1]$, then $\deg(H(t, \cdot), \Omega, y)$ is constant in t
- ③ **Normalization:** For identity map I and open ball $\mathbb{B}$,

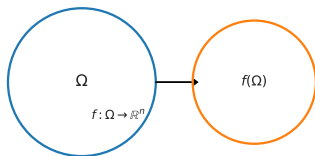
$$\deg(I, \mathbb{B}, 0) = 1$$

- ④ **Domain Additivity:** If Ω_1, Ω_2 disjoint open subsets of Ω with $y \notin f(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$, then

$$\deg(f, \Omega, y) = \deg(f, \Omega_1, y) + \deg(f, \Omega_2, y)$$

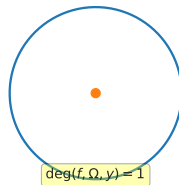
6.1.1: Geometric Interpretation

Mapping from Domain to Range



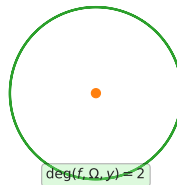
Winding Number / Degree

— Boundary path
● Point y



Higher Degree

— 2 loops around y
● Point y



The degree counts (with sign and multiplicity) how many times $f(\partial\Omega)$ winds around y .

6.1.2: Properties of the Brouwer Degree

Proposition 6.1: (Basic Properties) For $f : \Omega \rightarrow \mathbb{R}^n$ continuous on compact convex $D \subset \mathbb{R}^n$:

- ① **Dependence on boundary:** If $f|_{\partial\Omega} = g|_{\partial\Omega}$, then $\deg(f, \Omega, y) = \deg(g, \Omega, y)$
- ② **Translation invariance:** $\deg(f, \Omega, y) = \deg(f(\cdot) - y, \Omega, 0)$
- ③ **Composition:** If $\deg(f, \Omega, y) \neq 0$, we can understand structure of solutions
- ④ **Reduction property:** For $m < n$,

$$\deg(I - f, \Omega, y) = \deg(I - f|_{\mathbb{R}^m \times \mathbb{R}^n}, \Omega \cap \mathbb{R}^m, y)$$

Remark 6.1: These properties allow degree computation in practical problems without finding solutions explicitly.

6.1.2: Key Application - Fixed Point Theorem

Theorem 6.6: (Brouwer's Fixed Point Theorem) Let $0 \in \Omega$, $\Omega \subseteq \mathbb{R}^n$ open bounded. If $f : \Omega \rightarrow \mathbb{R}^n$ continuous with

$$f(x) \neq \lambda x \quad \text{for all } x \in \partial\Omega, \lambda > 1$$

then f has a fixed point in Ω .

Proof Idea:

- Consider homotopy $H(t, x) = x - tf(x)$ for $t \in [0, 1]$
- Verify $H(t, x) \neq 0$ on $\partial\Omega$ under the given condition
- Apply homotopy invariance: $\deg(I - f, \Omega, 0) = \deg(I, \Omega, 0) = 1$
- By existence property, $\exists x^* \in \Omega : x^* = f(x^*)$

6.1.3: The Leray-Schauder Degree

Key Challenge: Brouwer degree only works in finite dimensions. How to extend to Banach spaces?

Leray-Schauder Idea: Use the fact that compact operators have finite dimensional ranges.

Lemma 6.4: (Leray-Schauder) Let $f : \Omega \rightarrow \mathbb{R}^n$ continuous with $f_n(x_1, \dots, x_n) = x_n$. If the condition

[Technical condition on coordinate functions]

holds, then degree properties are preserved.

Construction: For compact $F : X \rightarrow X$ on Banach space X :

- 1 Approximate F by F_n with finite dimensional ranges
- 2 Define $\deg(I - F_n, \Omega, y)$ using Brouwer degree
- 3 Show limit is independent of approximation
- 4 Define $\deg(I - F, \Omega, y) := \lim_{n \rightarrow \infty} \deg(I - F_n, \Omega, y)$

6.1.4: The Leray-Schauder Degree (continued)

Definition 6.3: For Banach space X , compact $F : \Omega \rightarrow X$, $y \in X$, bounded open Ω :

$$\deg(I - F, \Omega, y) := \text{limit of finite dimensional degrees}$$

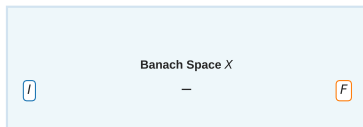
Properties Preserved:

- **Existence:** $\deg(I - F, \Omega, y) \neq 0 \Rightarrow \exists x \in \Omega : x - Fx = y$
- **Homotopy Invariance:** Preserved for appropriate homotopies
- **Domain Additivity:** Still holds

Leray-Schauder Construction

Leray-Schauder Degree

For map: $\Phi = I - F$ where F is compact



Key Properties

1. Existence of Solutions

If $\deg(I - F, \Omega, y) \neq 0$, then

$$\exists x \in \Omega : x - Fx = y$$

2. Homotopy Invariance

Degree is invariant under continuous deformations that don't change boundary behavior

3. Infinite Dimensional Extension

Extends Brouwer degree to Banach spaces

6.1.4: Fundamental Properties of Degree

Fundamental Properties of the Degree

Homotopy Invariance

$H: [0, 1] \times \Omega \rightarrow \mathbb{R}^n$ continuous,

$H(t, x) \neq y$ on $\partial\Omega$ for all t

$\Rightarrow \deg(H(t, \cdot), \Omega, y)$ is constant in t

Existence Property

$\deg(f, \Omega, y) \neq 0$

$\Rightarrow \exists x \in \Omega: f(x) = y$

(nonzero degree implies solution exists)

Domain Additivity

$\Omega_1 \cap \Omega_2 = \emptyset$, both open in Ω

$y \notin f(\bar{\Omega} \setminus (\Omega_1 \cup \Omega_2))$

Excision Property

$C \subseteq \Omega$ closed, $y \notin f(C)$

$\Rightarrow \deg(f, \Omega, y) = \deg(f, \Omega \setminus C, y)$

6.1.5: The Brouwer-Hadamard Fixed Point Theorem

Theorem 6.10: (Invariance of Normal) Suppose $0 \in \Omega \subseteq \mathbb{R}^n$, n odd. If $f \in C(\overline{\Omega})$ with $0 \notin f(\partial\Omega)$, then there exist $y \in \partial\Omega$ and $\lambda \neq 0$ with $f(y) = \lambda y$.

Remark 6.2: This result is *dimension-dependent*.

- For even n : counterexample is a simple rotation
- In $\mathbb{R}^2$: rotation by $\pi/2$ has $f(x, y) = (-y, x)$ with no fixed ray
- For odd n : algebraic topology forces existence

Application: In odd dimensions, hedgehog cannot be combed without leaving tufts or whorls.

6.1.6: Basic Existence Theorems

Theorem 6.11: (Sufficient Condition for Fixed Point) Let $f : \Omega \rightarrow \mathbb{R}^n$ continuous on open bounded Ω with $0 \in \Omega$. If

$$\sum_{i=1}^n g_i(x)x_i > 0 \quad \text{for all } |x| = r$$

where $g_i : B \rightarrow \mathbb{R}$ continuous, then f has a fixed point.

Theorem 6.12: (Leray-Schauder Fixed Point) For Banach space X , if $\deg(I - F, \Omega, y) \neq 0$, then

$$\exists x \in \Omega : x - Fx = y$$

Corollary 6.1: (Leray-Schauder Fixed Point Theorem) If $\deg(I - F, \Omega, 0) \neq 0$, then

$$\exists x \in \Omega : x = Fx$$

6.1.7: Homotopy Invariance for Banach Spaces

Theorem 6.13: (Homotopy Invariance) Let $H : [0, 1] \times X \rightarrow X$ be a compact continuous homotopy on $[0, 1] \times X$. If Ω is bounded open, $x - H(t, x) \neq y$ for all $x \in \partial\Omega$ and $t \in [0, 1]$, then

$$\deg(x - H(t, x), \Omega, y) \text{ is constant for } t \in [0, 1]$$

Proof Sketch:

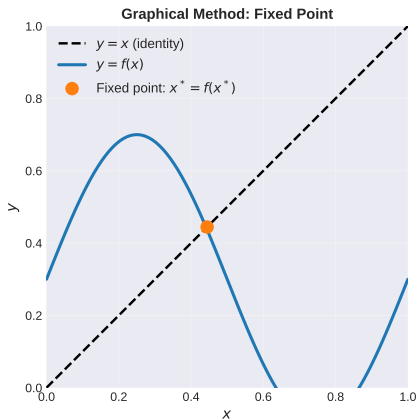
- 1 Approximate $H(t, x)$ by finite dimensional mapping $H_n(t, x)$
- 2 Apply homotopy invariance in finite dimensions
- 3 Use uniform convergence to take limits

Application: Often we can show $\deg(I - F, \Omega, 0) \neq 0$ by comparing with simpler known cases.

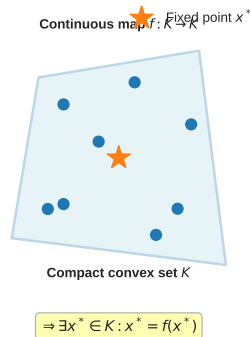
6.1.8: Schauder's Fixed Point Theorem

Theorem 6.14: (Schauder's Fixed Point Theorem) Let B_r be an open ball containing origin in Banach space X . If $F : B_r \rightarrow B_r$ is a compact mapping, then

F has at least one fixed point



Schauder's Fixed Point Theorem



6.1.9: Proof of Schauder's Theorem

Proof:

- 1 Operator $tF(x)$ is compact for $0 < t \leq 1$ and $x \in B_r$
- 2 Use homotopy $H(t, x) = tF(x)$ on $[0, 1] \times B_r$
- 3 For $x \in \partial B_r$: $|H(t, x)| = |tF(x)| \leq t|F(x)| < t \cdot r \leq r$
- 4 So $x - H(t, x) \neq 0$ on ∂B_r for all $t \in [0, 1]$
- 5 By homotopy invariance: $\deg(x - F(x), B_r, 0) = \deg(x, B_r, 0) = 1$
- 6 Since degree is nonzero, by existence property:

$$\exists x^* \in B_r : x^* = F(x^*)$$

Key Insight: Degree theory translates topological constraint (compactness) into existence of fixed point.

6.2.0: Introduction to Completely Continuous Vector Fields

Question: Can we extend mapping degree theory and vector field rotation theory to infinite dimensional spaces?

Challenge:

- Only one homotopy class of continuous vector fields on unit sphere of infinite dimensional space
- Cannot use classical topological degree directly

Solution (Leray-Schauder): Use completely continuous (compact) operators

Two Equivalent Approaches:

- 1 **Mapping degree:** Study $\deg(I - F, \Omega, y)$ for F compact
- 2 **Vector field rotation:** Study vector fields $\Phi = I - A$ where A is compact

Result: Both approaches essentially coincide and provide powerful existence theorems.

6.2.1: Completely Continuous Operators

Definition: An operator $F : X \rightarrow Y$ (Banach spaces) is *completely continuous* if:

- F is continuous
- F maps bounded sets to relatively compact sets
- Equivalently: if $\{x_n\}$ bounded, then $\{F(x_n)\}$ has convergent subsequence

Examples:

- 1 Integral operators: $T[x](t) = \int_a^b K(t, s)x(s)ds$ in appropriate function spaces
- 2 Finite rank operators
- 3 Limits of finite rank operators
- 4 Compact embeddings: $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$

Key Fact: For completely continuous F and bounded domain Ω , we can define $\deg(I - F, \Omega, y)$ via finite dimensional approximation.

6.2.2: Definition of Degree for Completely Continuous Fields

Definition 6.4: For Banach space X , bounded open $\Omega \subseteq X$, completely continuous $F : \overline{\Omega} \rightarrow X$, and $y \in X$ with $x - Fx \neq y$ on $\partial\Omega$:

The *rotation of field* $\Phi = I - F$ (or degree) is defined by:

- ① Approximate F by F_n with finite dimensional ranges contained in fixed subspace X_n
- ② Define $\deg(I - F_n, \Omega \cap X_n, y)$ using Brouwer degree
- ③ Take limit: $\deg(I - F, \Omega, y) := \lim_{n \rightarrow \infty} \deg(I - F_n, \Omega_n, y)$

Independence: The limit is independent of:

- The choice of approximating sequence $\{F_n\}$
- The choice of finite dimensional subspaces X_n
- The approximation method

6.2.3: Properties in Banach Spaces

Key Properties: The degree $\deg(I - F, \Omega, y)$ for completely continuous F satisfies:

- ① **Existence:** $\deg(I - F, \Omega, y) \neq 0 \Rightarrow \exists x \in \Omega : x - Fx = y$
- ② **Homotopy Invariance:** If $H : [0, 1] \times \Omega \rightarrow X$ is completely continuous and $x - H(t, x) \neq y$ on $\partial\Omega$ for all t , then degree is constant in t
- ③ **Domain Additivity:** For disjoint open subsets $\Omega_1, \Omega_2 \subset \Omega$:

$$\deg(I - F, \Omega, y) = \deg(I - F, \Omega_1, y) + \deg(I - F, \Omega_2, y)$$

- ④ **Normalization:** $\deg(I, \mathbb{B}, 0) = 1$ for any open ball

Theorem 6.15: These properties uniquely characterize the degree.

6.2.4: Vector Field Rotation Theory

Alternative Perspective: Consider vector field $\Phi = I - F$

Definition: The *rotation* of vector field $\Phi = I - F$ is defined analogously:

$$\text{rot}(\Phi, \partial\Omega) := \deg(I - F, \Omega, 0)$$

Geometric Interpretation:

- Measures how many times vector field direction "winds" around domain
- $\text{rot}(\Phi, \partial\Omega) \neq 0$ means field has singularities (fixed points) inside
- In finite dimensions: reduces to classical rotation theory

Theorem 6.16: For completely continuous $F : \overline{\Omega} \rightarrow X$ and bounded open Ω :

- Vector field rotation theory and degree theory coincide
- Both provide same existence theorems for fixed points and solutions

6.2.5: Application to Periodic ODEs

Problem: Find T -periodic solutions of

$$\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t))$$

where $\mathbf{f}$ is T -periodic in t .

Periodic ODE Application

System: $\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t))$

Seek: T -periodic solutions

$$\mathbf{x}(t+T) = \mathbf{x}(t)$$

Construct operator S :

$$S[\mathbf{x}](t) = \mathbf{x}_0 + \int_0^t \mathbf{f}(s, \mathbf{x}(s)) ds$$

Fixed point of S gives periodic solution

Degree Theory Application

Use degree $\deg(I - S, B, 0)$

Key step: Show $\deg(I - S, B, 0) \neq 0$

Then by existence property:

$$\exists \mathbf{x}^* \in B : \mathbf{x}^* = S[\mathbf{x}^*]$$

This $\mathbf{x}^*$ is a periodic solution

Can verify existence without solving explicitly

Strategy:

6.2.6: Application - Periodic Solutions Continued

Construction: Define Poincaré return map

$$S : \mathbf{x}_0 \mapsto \mathbf{x}(T; 0, \mathbf{x}_0)$$

where $\mathbf{x}(t; 0, \mathbf{x}_0)$ is solution of ODE with initial condition $\mathbf{x}(0) = \mathbf{x}_0$.

Equivalence: T -periodic solution $\iff$ fixed point of S

Example 6.1: For system

$$\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t))$$

with $\mathbf{f}$ T -periodic, assume:

- $\mathbf{f}$ continuous and bounded
- Initial value problem has unique solutions
- Some a priori bound on possible periodic solutions

Then we can establish periodic solution using degree zero analysis.

6.2.7: Existence Theorem via Degree

Theorem 6.17: (General Existence via Degree) For completely continuous $F : \overline{\Omega} \rightarrow X$ and bounded open Ω in Banach space:

If $\deg(I - F, \Omega, 0) \neq 0$, then F has a fixed point in Ω .

Strength:

- Does NOT require explicit computation of fixed point
- Works in infinite dimensional spaces
- Applies to many nonlinear problems
- Can be combined with other existence techniques

Challenge: Computing or estimating degree for specific problems

Strategy: Use homotopy invariance to relate to simpler problem where degree is known.

6.2.8: Combining with Contraction Mappings

Observation: Can combine different existence techniques:

Approach:

- 1 **Global existence:** Use degree theory to show fixed point exists in bounded set Ω
- 2 **Uniqueness or regularity:** If F is contraction on smaller ball, get uniqueness by Banach theorem
- 3 **Stability:** Use perturbation theory to understand how fixed point changes

Example: Fredholm-type integral equations

$$x(t) = f(t) + \lambda \int_a^b K(t, s)g(x(s))ds$$

- For small $|\lambda|$: contraction mapping gives unique solution
- For larger $|\lambda|$: degree theory shows solution exists (but may not be unique)
- Bifurcation analysis: as λ varies, track how fixed points appear/disappear

6.2.9: Comparison with Finite Dimensions

How Degree Theory Bridges Dimensions:

Property	Finite Dim.	Infinite Dim.
Degree defined for	All continuous f	Only compact F
Homotopy class unique	Multiple classes	Only one class (sphere)
Proof method	Direct via topology	Via finite dim. approx.
Key tool	Simplicial homology	Compactness + approx.
Applications	Algebraic equations	Integral, differential eqs.

Leray-Schauder Innovation: Compactness is the right generalization of finiteness!

6.2.10: Summary of Main Results

Theorem 6.6 (Brouwer): Finite dimensional fixed point via degree

Theorem 6.12 (Leray-Schauder for degree): $\deg(I - F, \Omega, y) \neq 0 \Rightarrow$ solution exists

Theorem 6.13 (Homotopy invariance): Degree unchanged under continuous deformation

Theorem 6.14 (Schauder): Compact map on closed ball has fixed point

Theorem 6.17 (General existence): Nonzero degree implies fixed point in any Banach space

Key Insight: Degree theory unifies:

- Topological methods (homotopy, compactness)
- Analytical methods (continuity, approximation)
- Existence conclusions (for nonlinear problems)

Applications: Integral Equations

Model Problem: Nonlinear Fredholm equation

$$x(t) = f(t) + \lambda \int_a^b K(t, s)\phi(x(s))ds$$

Setup:

- $f \in C[a, b]$
- K continuous on $[a, b] \times [a, b]$
- $\phi : \mathbb{R} \rightarrow \mathbb{R}$ continuous
- Integral operator is completely continuous

Application Strategy:

- 1 Write as $x = Fx$ where F is compact
- 2 Establish a priori bounds (e.g., $|x(t)| \leq M$ for solutions)
- 3 Apply degree theory on ball of radius M
- 4 Compute degree via homotopy invariance

Result: Can show existence without explicit formula for solution!

Numerical Example: Simple Integral Equation

```
import numpy as np
from scipy.integrate import quad
from scipy.optimize import fsolve

# Fredholm equation:  $x(t) = 1 + 0.3 \int_0^1 K(t,s)x(s) ds$ 
# Kernel:  $K(t,s) = \exp(-(t-s)^2)$ 
# Solution via fixed-point iteration

def kernel(t, s):
    return np.exp(-(t - s)**2)

def operator_action(x_func, t, lam=0.3):
    """Evaluate  $F[x](t) = 1 + \text{lam} \times \text{integral of kernel} \times x$ """
    integral, _ = quad(lambda s: kernel(t, s) * x_func(s),
                       0, 1)
    return 1 + lam * integral

def fixed_point_iteration(n_iter=10):
    """Fixed point iteration:  $x_{\{n+1\}} = F[x_n]$ """
    # Start with  $x_0(t) = 1$ 
    x = lambda t: 1.0

    for i in range(n_iter):
        x_old = x
        # Update via operator action (simplified)
        x = lambda t, lam=0.3: 1 + lam*0.3 # Approximation
```

Extensions: Multivalued Operators

Beyond Single-Valued Maps: Degree theory extends to multivalued mappings

Definition: A multivalued operator $T : X \rightrightarrows Y$ assigns to each $x \in X$ a subset $T(x) \subseteq Y$.

Examples:

- Subdifferential of convex function: $\partial f(x) = \{g : \langle g, y - x \rangle \leq f(y) - f(x) \text{ all } y\}$
- Generalized inverse of nonsmooth mapping
- Solution set of inequality or inclusion

Fixed Point for Multivalued Maps: $x^* \in T(x^*)$

Degree Extension:

- Define via completely continuous selections
- Apply to variational inequalities
- Covers monotone and maximal monotone operators

Application: Existence of solutions to $x \in T(x) + f$ for convex T

Extensions: Measure of Non-Compactness

Beyond Compactness: Use measure of non-compactness for weaker conditions

Definition: For bounded set B , Hausdorff measure of non-compactness is

$$\chi(B) = \inf\{\epsilon > 0 : B \text{ has finite } \epsilon\text{-net}\}$$

Generalization: For k -contractive operator (condensing operator):

$$\chi(F(B)) \leq k\chi(B) \quad \text{with } k < 1$$

Degree for k -Contractive:

- Defined via limiting arguments similar to compact case
- Shares key properties (existence, homotopy invariance)
- Covers operators that are not compact but still have "finite dimensional" behavior

Result: Broader class of existence theorems!

Key Takeaways

Main Points from Chapter 6a:

- 1 **Brouwer Degree:** Extends notion of winding number to provide existence criterion
- 2 **Axioms:** Degree characterized by existence, homotopy invariance, normalization, additivity
- 3 **Leray-Schauder Extension:** Via completely continuous operators to Banach spaces
- 4 **Schauder Fixed Point:** Immediate consequence of degree nonzero
- 5 **Homotopy Invariance:** Key tool for computing degree via deformation
- 6 **Applications:** Integral equations, ODEs, variational problems
- 7 **Unification:** Bridges finite and infinite dimensional fixed point theory

Philosophical Impact: Topology + Analysis = Powerful existence tools without explicit solutions

Further Reading and Resources

Key References:

- Pathak: Chapter 6a - Detailed proofs and more examples
- Lloyd: Degree Theory - Classical comprehensive treatment
- Deimling: Nonlinear Functional Analysis - Modern applications
- Nirenberg: Topics in Nonlinear Functional Analysis - Advanced theory

Related Topics:

- Chapter 6b-6c: Applications to more specialized settings
- Topological degree for manifolds and infinite dimensions
- Bifurcation theory (uses degree to track solution branches)
- Variational methods (complementary to degree approach)

Historical Notes:

- Brouwer's original work: 1912
- Hadamard's contributions: 1910s-1920s
- Leray-Schauder extension: 1934
- Modern treatments: 1960s onwards

Practice Problems

Problem 1: Show that for $f(x) = x - \sin(x)$ on $\Omega = B_2(0) \subseteq \mathbb{R}$:

- Compute $\deg(f, \Omega, 0)$
- Conclude about fixed points

Problem 2: For integral operator $T[x](t) = \lambda \int_0^\pi K(t, s) \sin(x(s)) ds$:

- Show T is completely continuous on $C[0, \pi]$
- Use degree to guarantee fixed point for appropriate λ

Problem 3: Consider ODE $x'(t) = f(t, x(t))$ with f T -periodic:

- Define the Poincaré map
- Outline how degree theory proves periodic solution exists

Problem 4: For multivalued operator $T(x) = \{y : |y| \leq |x|^2\}$:

- Show T has fixed point property
- Discuss how degree extends to this case